Olive oil (virgin, refined, pomace): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

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Abstract

The Heavy Metal Tested & Certified standard for Olive oil (virgin, refined, pomace) (Culinary Oils and Fats), stated as sold. Each analyte carries the strictest applicable government maximum level in the product's native basis, with a disclosed read-across or a mandatory control where no government regulates the exact product.

1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; face value, no reconstitution; read-across / control / not-material for unregulated cells; no occurrence percentiles)

2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

ANALYTE	TIER	HMTC LIMIT	BASIS
Lead (Pb)	Tier 1	80	government maximum level
Cadmium (Cd)	Tier 1	30	government maximum level
Total arsenic (tAs)	Tier 2	100	government maximum level
Mercury, total (tHg)	Tier 1	20	government maximum level

Nickel (Ni)	Tier 2	100	human-grade read-across (European Union)
Tin (Sn)	Tier 2	200,000	human-grade read-across (European Union)
Aluminium (Al)	Tier 2	1,000	human-grade read-across (European Union)
Total chromium (Cr)	Tier 2	300	human-grade read-across (China)
Hexavalent chromium (Cr-VI)	Tier 2	10	human-grade read-across (US-CA)

3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

ANALYTE	HMTC	SOVEREIGN ANALOG / DETERMINATION
Lead (Pb)	80	government maximum level
Cadmium (Cd)	30	government maximum level
Total arsenic (tAs)	100	government maximum level
Inorganic arsenic (iAs)	100	Inorganic arsenic is a component of total arsenic (iAs <= tAs). The iAs value (100 ug/kg) is not below the binding total arsenic ceiling (100 ug/kg), so inorganic arsenic is governed by and bounded by the total arsenic ceiling; no separate iAs ceiling is published. It is enforced by reflex speciation whenever tAs is elevated.
Mercury, total (tHg)	20	government maximum level
Methylmercury (MeHg)	20	Methylmercury is a component of total mercury (MeHg <= tHg). The MeHg value (20 ug/kg) is not below the binding total mercury ceiling (20 ug/kg), so methylmercury is governed by and bounded by the total mercury ceiling; no separate MeHg ceiling is published. It is enforced by reflex speciation whenever tHg is elevated.

Nickel (Ni)	100	European Union — human-food analog
Tin (Sn)	200,000	European Union — human-food analog
Aluminium (Al)	1,000	European Union — human-food analog
Total chromium (Cr)	300	China — human-food analog
Hexavalent chromium (Cr-VI)	10	US-CA — human-food analog

4 | Cite this standard

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References

[1] China — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] European Union — human-food maximum levels
Binds Ni, Sn, Al. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] US-CA — human-food maximum levels
Binds Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

STATUS AND LICENCE

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COMPETING INTERESTS

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DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/olive-oil. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/olive-oil.